Motherboard shootout
88 19 Z77 motherboards duke it out in this month's motherboard comparison. Which is the right one for you?

Budget smart phones
94 On a tight budget but still on the lookout for a smart phone? We give you the lowdown on 8 such phones

Wireless Warriors

We put 20 routers through their paces. Read on to find out which one is the right pick for you and also get some pointers on setting up a home network

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The sheer number of wireless devices at our disposal is staggering. That, coupled with our data consumption rate means that the bandwidth we require has gone through the roof. Gone are the days of DVD video and just checking your mail on your phone and PC, we want to watch HD movies, browse YouTube on the go, indulge in multiplayer gaming, etc.

It doesn't stop there, these same wireless devices have also managed to clutter up the airwaves. Most of them operate in the 2.4 GHz band and will interfere with your network. Because of this, we're slowly seeing a shift to dual-band routers. These routers can stream on both the 2.4 GHz and 5 GHz band, which, while suffering less interference, also results in a drastic drop in range and penetrative power. Also, your receiver needs to be capable of receiving that 5 GHz signal.

There is a lot more to routers than just buying a device that looks "nice"or claims or to have a lot of features though. You need to consider your usage, the number of wireless devices that you want to use, your RF (radio frequency) environment, device positioning, receiver capabilities and numerous other criteria. During this test we learned many things ourselves and will share that knowledge with you and help you to choose the correct router as well as to setup a wireless network.

SETTING UP A NETWORK

The first and most important step in setting up your Wi-Fi network is determining what sort of receiver you have. Just because you have a 600N dual-band router doesn't mean that your receiver can actually receive that signal. Check up the model number of your receiver online or in your user manual, etc. to determine the capability of your device. You might see 2x2 or 1x2, etc. written alongside your receiver specs, these numbers indicate the number of transmit and receive antennas that your adapter uses respectively. Usually, the more antennas the adapter has, the higher the total data throughput. Ensure that you count antennas per band as this will determine your actual data throughput per band. Simply put, a 1x2 adapter can receive better than it can transmit, which is perfectly fine if you want to stream data to your device, rather than from it. As an example, a media server requires better transmission, an HDTV requires better reception, setup your systems accordingly.

Location is another important factor to consider when setting up your router and